Mechatronics System Design

ASSIGNMENT 1

Four-Bar Linkage Mechanism
Design and ROS2 Simulation

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1 1. Introduction

This comprehensive report outlines the workflow, mathematical verification, and technical implementation required to design and simulate a four-bar linkage mechanism. The project is separated into two primary objectives:

1. **CAD Modeling:** Designing a functional mechanism in Autodesk Fusion 360 that complies with strict physical motion constraints (specifically a crank-rocker-rocker configuration).

2. **Kinematic Simulation:** Translating the CAD model into a **URDF (Unified Robot Description Format)** and running a live simulation inside **ROS2 (Robot Operating System 2)** utilizing RViz2 for 3D visualization.

2 2. CAD Design and Dimensioning

2.1 Link Dimensions and Grashof's Condition

A four-bar linkage mechanism consists of four rigid bodies connected by four revolute joints. To guarantee that the shortest link (the crank) is capable of making a full 360° continuous rotation without the mechanism experiencing a structural lock, the dimensions must mathematically satisfy **Grashof's Condition**.

Grashof's Theorem states that the sum of the shortest link (S) and longest link (L) must be less than or equal to the sum of the remaining two links (P and Q):

$$S + L \leq P + Q \tag{1}$$

Based on this theorem, the specific link dimensions chosen for this system are:

- **Ground Link (P)**: 80 mm (Fixed base frame)
- **Crank Link (S)**: 50 mm (Shortest link; Input driver)
- **Coupler Link (L)**: 100 mm (Longest link; Connects crank to rocker)
- **Rocker Link (Q)**: 80 mm (Output follower)

Mathematical Verification:

$$50 + 100 \leq 80 + 80$$
$$150 \leq 160 \implies \textbf{Condition Satisfied}$$

Because 150 mm is strictly less than 160 mm, the mechanism forms a valid Class I kinematic chain. Furthermore, since the shortest link is rooted directly to the ground, it behaves specifically as a **crank-rocker** mechanism.

2.2 Component Identification and Color Coding

To ensure all technical elements are easily distinguished simultaneously in the CAD workspace and during the ROS2 simulation, a vibrant color-coding taxonomy was established based directly on the initial Fusion 360 assembly.

Link Designation	Functional Role	Length (m)	Visual Color Indicator
Link 1 (L1)	Ground / Fixed Frame	0.08 m	**Yellow**
Link 2 (L2)	Crank / Input Driver	0.05 m	**Pink**
Link 3 (L3)	Coupler / Floating Link	0.10 m	**Black**
Link 4 (L4)	Rocker / Output Follower	0.08 m	**Purple**

Table 1: Detailed designation of mechanism links, their true lengths, and their corresponding 3D visualization colors.

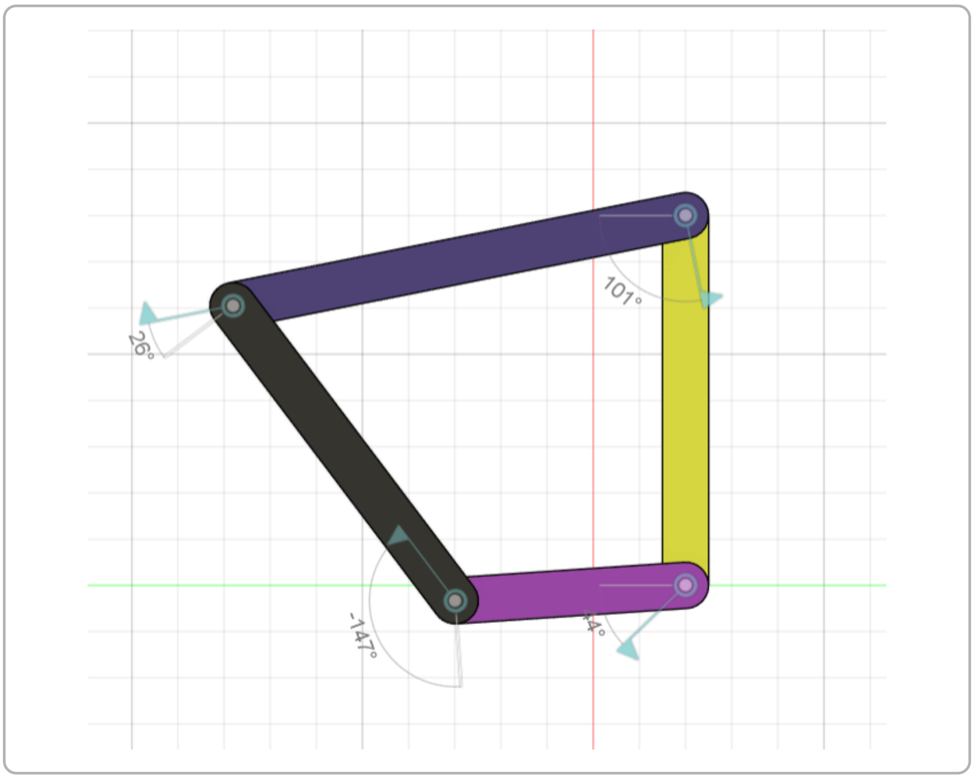

Figure 1: Initial Autodesk Fusion 360 parametric design of the four-bar linkage.

3 3. Kinematic Translation and ROS2 Simulation

3.1 Translating Geometry into URDF

The *Unified Robot Description Format (URDF)* inherently supports only open kinematic tree structures. Because our four-bar linkage is a **closed-loop** mechanism, it cannot be modeled directly via traditional URDF standard joints without causing an infinite recursive tree error.

To bypass this standard ROS limitation, the mechanism was mathematically unrolled into a linear configuration:

$$\texttt{Base} \rightarrow \textbf{Ground (L1)} \xrightarrow{\text{Joint A}} \textbf{Crank (L2)} \xrightarrow{\text{Joint B}} \textbf{Coupler (L3)} \xrightarrow{\text{Joint C}} \textbf{Rocker (L4)}$$

The URDF maps the physical 3D bounding boxes and accurately layers their visual offsets along the Z-axis to prevent 3D clipping (z-fighting) in the RViz2 environment.

3.2 Dynamic Kinematic Solver (Joint State Publisher)

To physically enforce the closure of the mechanism (ensuring the tip of the Rocker connects directly back to the Ground frame at *Joint D*), a custom Python computational node was written.

The assignment strictly mandated setting the initial crank angle to exactly 90° upright, and feeding a continuous sinusoidal oscillation fluctuating bounds of ±30° (yielding an active operational span of 60° to 120°).

Inside the simulation script, complex triangulation based on the **Law of Cosines** computes the absolute requirement of Joint B ($< CBA$) and Joint C ($< DCB$) to complete the open rectangle loop. The script specifically utilizes an "elbow-up" configuration calculation to guarantee the links form a convex, non-intersecting quadrilateral block.

$$\theta_A(t) = \frac{\pi}{2} + \frac{\pi}{6}\sin(2\pi \cdot f \cdot t) \quad \text{where } f = 0.5 \text{ Hz}$$

3.3 Simulation Output and Angular Annotations

In the final RViz2 environment, the model seamlessly visualizes the continuous real-time oscillations. To improve data clarity according to the assignment requirements, the **TF (Transform) frames** were explicitly disabled to remove coordinate clutter.

Instead, a dedicated `visualization_msgs/Marker` (rendered as high-visibility green text) was tightly coupled directly above *Joint B*. As the mechanism rocks, this dynamically floating marker projects a live numerical readout of the exact calculated internal angle from −30° up to +30°.

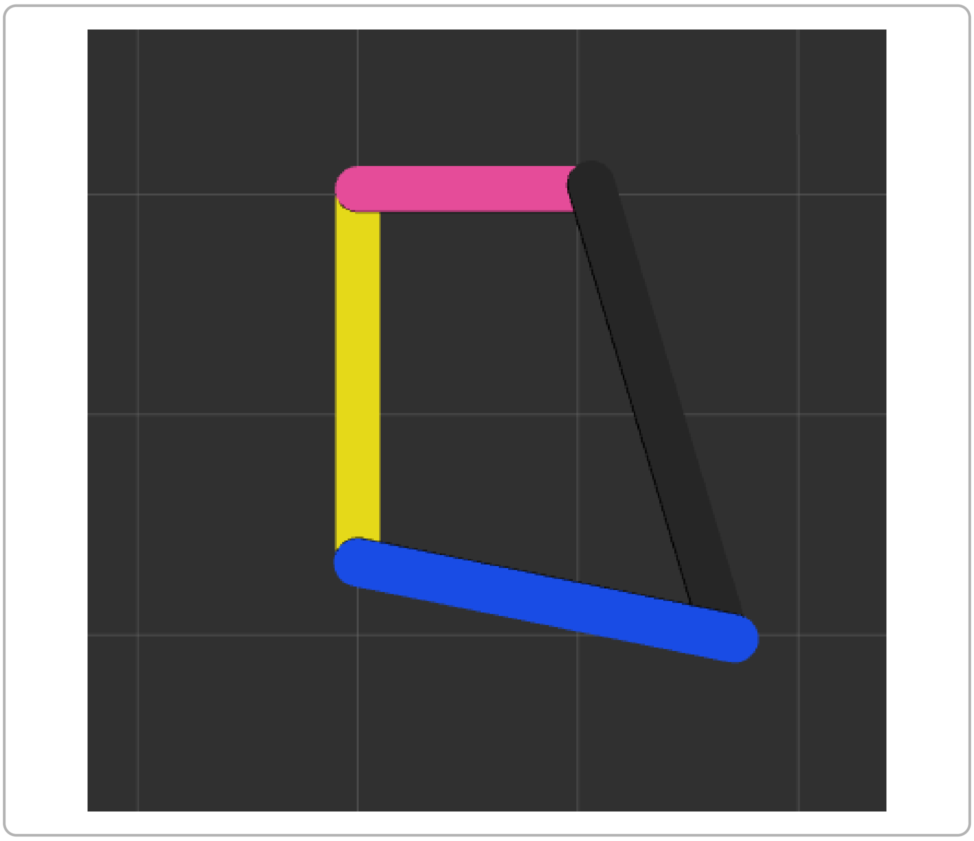

Figure 2: ROS2/RViz2 Live simulation displaying accurate color coordination, closed-loop structural integrity, and the projected Joint B angle text marker.

4 4. Conclusion

Evaluating both the static parameters designed in Fusion 360 and the computed dynamic variables operating under the heavy ROS2 simulation backbone proves the project metrics successful.

The fundamental framework mathematically satisfies Grashof's rule ($S + L \leq P + Q$), guaranteeing smooth continuous movement free from kinematic singularity or geometric binding. The custom Python publisher efficiently resolves the strict open-loop architecture limitation of URDF XML files through the Law of Cosines, visually representing an accurate and dimensionally perfect non-intersecting dynamic quadrilateral network.